

Oswald Gyabaah

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Accra, Ghana

PROFESSIONAL SUMMARY

AI engineer and infrastructure builder shipping production LLM systems and treating prompt engineering as a primary craft. Build and maintain multi-mode prompt frameworks with explicit system-prompt and user-context separation, RAG retrieval over local knowledge bases, few-shot scoring rubrics, and multi-agent orchestration with tool-use. Production AI work includes Palsar (Claude API monitoring SaaS), Root-ID (AI agent identity platform), career-ops (multi-mode prompt pipeline used to evaluate this role), and live R&D agents on Linux production servers. Founded Guaranty Trust Bank Ghana's DevOps practice from zero and led 12-person cross-functional teams as founding CTO of MyHealthCop, shipping the assess-tailor-measure training loop across non-engineering roles. Python depth across Django, FastAPI, Celery; deep Kubernetes and multi-cloud operations.

CORE COMPETENCIES

Prompt engineering - Chain-of-Thought, few-shot, RAG · Multi-agent orchestration and tool-use patterns · LLM API integration (Claude, OpenAI, Grok) · Prompt frameworks, templates, and shared prompt libraries · AI use case discovery and cross-functional facilitation · Training and upskilling - assess, tailor, measure · Python automation (Django, FastAPI, Celery) · Production AI systems with structured outputs and observability

PROFESSIONAL CERTIFICATIONS

AI Engineering Specialization · *ByteByteAI* · Mar 2026
Oracle Certified Architect Associate · *Oracle* · Aug 2025
Oracle Certified DevOps Professional · *Oracle* · Aug 2025
Oracle Certified Foundations Associate · *Oracle* · Jul 2025
AWS Certified Cloud Practitioner · *AWS* · May 2024
Django Web Framework · *Coursera* · Dec 2022

TECHNICAL SKILLS

AI & Prompt Engineering: LLM API integration (Claude, OpenAI, Grok), prompt engineering (Chain-of-Thought, few-shot, RAG), multi-agent orchestration, tool-use patterns, prompt frameworks and shared libraries, structured outputs and prompt evaluation, agent design for QA and sysadmin operations
Languages: Python (Django, FastAPI, Celery), JavaScript, TypeScript, Bash, Node.js (Express)
Frontend: React.js, Next.js, Tailwind CSS, Flutter
Cloud & Infrastructure: AWS (EKS, ECS, Lambda, S3, SQS/SNS, EventBridge, DynamoDB, VPC), GCP (GKE, Cloud SQL), Azure, OCI, on-prem bare metal
Containers & Platforms: Docker, Kubernetes (EKS, GKE, TKG/Tanzu, RKE2), Helm, FluxCD, ArgoCD
Integration & APIs: REST APIs, GraphQL, WebSockets, Kafka, RabbitMQ, AWS SQS/SNS, EventBridge, SMS and email gateways, payment provider integrations
Databases: PostgreSQL, MySQL, MongoDB, Firebase/Firestore, DynamoDB, Convex, SQLite, Redis
Observability & CI/CD: Prometheus, Grafana, ELK Stack, Sentry, Terraform, Ansible, GitHub Actions, Azure DevOps
Leadership: Cross-functional team leadership (12+), team founding from zero, training and upskilling, stakeholder management, technical project management

Core Banking Services Lead · *Guaranty Trust Bank Ghana LTD* · Accra, Ghana May 2025 - Present

- Use AI agents and prompt-engineered workflows to accelerate L2 incident triage and migration test design - coordinated 10,000+ test cases across 13 core banking modules with AI-assisted documentation and runbook generation.
- Led a 5-person QA team through the Basis-to-Finacle migration cutover over a two-week window, designing and executing 10,000+ test cases across 13 modules and 33 internal applications - assessed each tester's baseline, tailored ramp-up paths, and measured outcomes daily.
- Own the on-prem Finacle core banking deployment on Tanzu Kubernetes Grid, managing deployments, patching, and incident response across 800+ production pods sustaining 1.5M+ daily transactions.
- Architected integrations between Finacle core banking and multiple third-party systems including transaction portals and internal tools, operating inside the Bank of Ghana regulatory perimeter.
- Built a bank-wide OTP service (Django + Next.js) integrating SMS and email gateways to secure core banking access for 600+ internal users, with structured logging to Elasticsearch via Filebeat for audit.
- Built a national ID verification service (Django + Next.js) processing and storing ID documents in self-hosted MinIO with automated verification workflows.
- Lead L2 support operations correlating logs across internal apps and core banking microservices via the Elastic Stack for root-cause analysis and remediation.

DevOps Engineering Lead · *Guaranty Trust Bank Ghana LTD* · Accra, Ghana Apr 2024 - Present

- Founded the bank's DevOps team of 4 engineers from zero - assessed each engineer's baseline (Git, Linux, Kubernetes), designed tailored ramp-up paths to a shared GitOps-native end state, ran weekly knowledge-share sessions and pair-programming rotations. All 4 productive on Kubernetes within 90 days.
- Architected high-availability Kubernetes clusters across multiple on-prem environments running internal applications handling 3M+ daily requests at 99.97% measured uptime.
- Implemented CI/CD automation that cut deployment time by ~90% and increased release velocity ~5x; introduced GitOps with FluxCD for declarative, version-controlled cluster deployments.
- Embedded Veracode and Trivy scanning into release pipelines as automated security gates; introduced declarative cluster config for rapid disaster recovery rebuilds.
- Automated 95% of infrastructure configuration management using Terraform, Ansible, Python, and Bash, turning multi-day manual processes into reproducible, version-controlled deployments.
- Managed multi-cloud infrastructure across Azure and AWS: resource provisioning, private VPC peering to on-prem systems, IAM, access control, and cost optimization.
- Introduced a unified observability stack (Prometheus, Grafana, Elastic Stack, Sentry) for centralized logging, tracing, and real-time alerting.

Chief Technical Officer (Founding) · *MYHEALTHCOP LTD* · Accra, Ghana Jan 2021 - Mar 2024

- Led a 12-person cross-functional team across product, design, QA, mobile, frontend, and backend through the full Agile lifecycle - designed sprint planning and design reviews for mixed-technical literacy, ran onboarding sessions for non-engineering hires on technical workflows, translated business requirements into architecture decisions for non-developer stakeholders.
- Architected the myhealthcop.com platform end-to-end across requirements gathering, system design, database modeling, and API development, delivering three applications: a patient-facing Flutter mobile app, a Flutter app for health professionals, and a React admin web dashboard.
- Designed AWS infrastructure running containerized microservices on EKS and ECS alongside Lambda functions for event-driven workloads, with autoscaling, load balancing, and IAM-controlled inter-service communication.
- Built a high-throughput background processing system handling 1M+ daily jobs across payments, messaging, and notifications using SQS/SNS with multi-provider failover.
- Led performance optimizations that reduced peak request latency by 90% through Redis caching, query tuning, and indexing in PostgreSQL.
- Designed a disaster recovery strategy mirroring AWS production on GCP using GKE, Cloud SQL, and Cloud Storage with secure VPCs and IAM policies.

- Automated application delivery and infrastructure provisioning via CI/CD pipelines on GitHub Actions and Terraform, making deployments repeatable, auditable, and reversible.

PROJECTS

AI Agents for Infrastructure Operations & career-ops Prompt Pipeline · *Personal R&D*

2024 - Present

- Build and maintain career-ops, a multi-mode AI prompt pipeline used daily for job-search workflows - explicit system-prompt + user-context separation, per-mode contracts, few-shot scoring rubrics, RAG retrieval over local knowledge bases, structured outputs validated against schemas. Used to evaluate this DLP role and generate this CV.
- Build specialized AI agents for QA automation, code review, and sysadmin operations running against live Linux production servers with real workloads - designed prompt frameworks and multi-agent orchestration patterns with tool-use.
- Maintain a documented, versioned prompt library (modes/, batch prompts, agent skill files) with clear conventions for system-layer vs user-layer customization - the same separation pattern that scales to team-wide prompt libraries.
- Built AWS Lambda functions triggered by EventBridge to deliver notifications on codebase events; used DynamoDB for NoSQL data modeling with both single-table and multi-table design patterns.

Stack: Claude API, OpenAI API, Python, Node.js, RAG patterns, multi-agent orchestration, AWS Lambda, EventBridge, DynamoDB

Palsar - AI-Powered Monitoring SaaS · *Founder*

2024 - Present

- Designed prompt-engineered workflows producing structured outputs at scale - explicit output schemas, few-shot examples calibrated to monitoring data patterns, retry-on-malformed-output logic, model-version pinning, prompt caching for cost control.
- Built downstream tool-use integrations triggered by LLM decisions; production AI monitoring system with stable structured outputs.

Stack: Next.js, TypeScript, Convex, Claude API, prompt caching, structured outputs

Virk Cloud (virk.cloud) - Platform-as-a-Service · *Founder & CTO*

Aug 2025 - Present

- Architected a Platform-as-a-Service for startups with built-in CI/CD, observability, and log streaming on bare-metal VPS clusters; built a management console for workload management, custom domains, and monitoring dashboards.

Stack: Next.js, TypeScript, Convex, Tailwind CSS, Kubernetes (RKE2), FluxCD, Terraform, Prometheus, Grafana, Python

EDUCATION

BSc Computer Engineering · *Ashesi University · Berekuso, Ghana*

Sep 2016 - May 2020